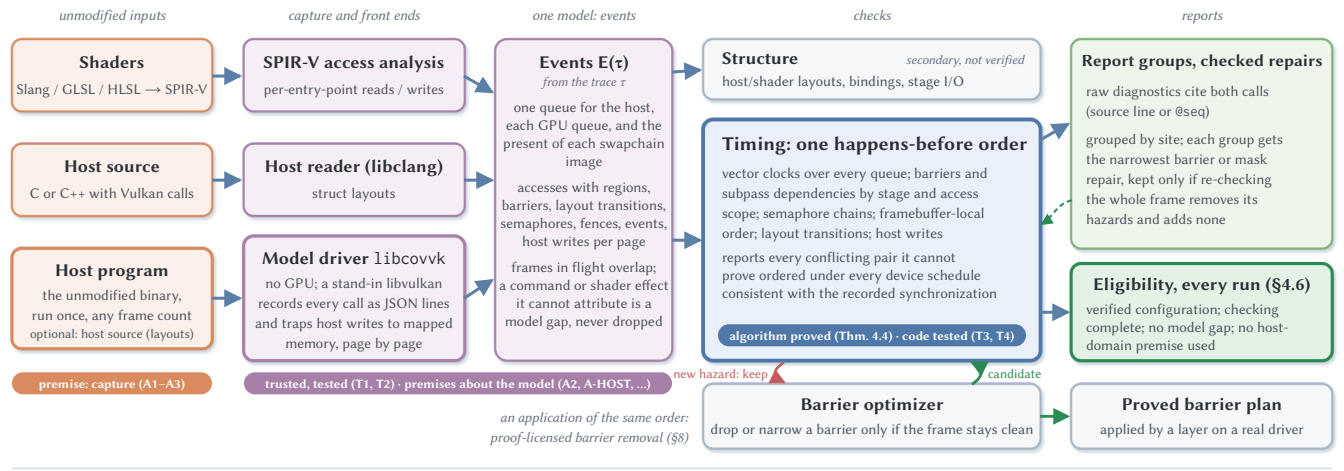


What a clean, eligible report rests on *each link is a premise, trusted and tested code, a proof, or a check on every run (Tables 2 and 4)*

1 Real execution
one run of the unmodified program; every call and host write recorded
premise about capture

2 Event extraction
trace and shaders become the events of Definition 4.1; gaps are reported, not dropped
trusted, tested; model premises

3 Abstract checker
reports every conflicting pair that the defined happens-before order leaves unordered
proved by hand (Theorem 4.4)

4 Eligibility
the run used the verified configuration and no gap or host-domain premise
checked and reported per run

No report ⇒ race-free
under every device schedule consistent with the recorded synchronization (Cor. 4.5), on the recorded host path

Beyond the recorded frames: Theorem 4.11 extends a clean result to later frames when the program repeats its period (premise P-PER, which a trace cannot establish).